

22593

12425

3 Hours / 70 Marks

Seat No.

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- Instructions :**
- (1) All Questions are *compulsory*.
 - (2) Answer each next main Question on a new page.
 - (3) Illustrate your answers with neat sketches wherever necessary.
 - (4) Assume suitable data, if necessary.

Marks

1. Attempt any FIVE of the following :

10

- (a) Write the applications of AI.
- (b) List different types of AI agent.
- (c) Define heuristic search technique.
- (d) Define the first order logic.
- (e) Define training and testing data.
- (f) State any two important supervised machine learning algorithms.
- (g) What is the need of confusion matrix ?

2. Attempt any THREE of the following :

12

- (a) Explain the need of data preprocessing.
- (b) Explain binary and multiclass classification in logistic regression.
- (c) Explain the different techniques of knowledge based agent in AI.
- (d) Write algorithm of depth first search.



- 3. Attempt any THREE of the following : 12**
- (a) Explain positive and negative class, cross validation.
 - (b) Explain turning test in AI with an example.
 - (c) Describe different metrics for classification.
 - (d) Explain the working of Hill Climbing algorithm in AI.
- 4. Attempt any THREE of the following : 12**
- (a) Describe structure of Agents.
 - (b) Describe the properties of A* algorithm.
 - (c) What are the techniques of knowledge representation ?
 - (d) Implement simple linear regression algorithm in Python.
- 5. Attempt any TWO of the following : 12**
- (a) Write a short note on : Supervised learning algorithm, unsupervised learning, semi-supervised learning algorithms.
 - (b) Describe machine learning life cycle.
 - (c) Elaborate beyond classical search.
- 6. Attempt any TWO of the following : 12**
- (a) Illustrate different type of reasoning in AI.
 - (b) Explain any one unsupervised learning algorithms.
 - (c) Explain following metrics for regression :
 - (i) Mean Squared Error (MSE)
 - (ii) Root Mean Squared Error (RMSE)
 - (iii) Mean Absolute Error (MAE)
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